LAUKIK BHALCHANDRA NAKHWA

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EDUCATION

National University of Singapore — Singapore
Research Scholar — August 2024 - Present

Birla Institute of Technology and Science, Pilani — Goa, India
Bachelor of Engineering in Electronics and Instrumentation Engineering — Oct 2020 - June 2025
Master of Science in Biological Sciences

SKILLS

Languages & Frameworks	Python (Expert), C/C++, SQL, FastAPI, PyTorch, TensorFlow, ROS, OpenCV
MLOps & Cloud	Docker, Kubernetes, Snowflake, Amazon S3, Dataiku (DSS), MLflow
Tools & Simulation	PyBullet, IsaacGym, MuJoCo, Git, MATLAB, Cadence Virtuoso

EXPERIENCE

Advanced Data Science Associate — December 2024 - Present
ZS Associates — *Gurgaon, India*

- **Global SCM Impact:** Developed and productionalized XGBoost and KNN models for **Pfizer's worldwide supply chain** to predict critical parameters (Lead-time, Shelf-life), optimizing global inventory distribution.
- **MLOps Transformation:** Engineered a modular, configuration-driven pipeline that reduced model deployment latency from **4 hours to 5 minutes** (98% efficiency gain) using **FastAPI** and **Kubernetes** orchestration.
- **Production Reliability:** Architected a specialized library for **3-tier Drift Detection** and automated data quality checks, ensuring 99.9% prediction reliability for large-scale, multi-region datasets.
- **Scalable Data Architecture:** Designed end-to-end ETL and feature engineering pipelines using **Snowflake and Amazon S3**, processing large-scale global data to support high-throughput production environments.
- **Industrial Experimentation:** Bridged the research-to-production gap by utilizing **Dataiku (DSS)** for rapid validation of SCM hypotheses before refactoring into scalable, industrial-grade Python modules.

Research Intern — July 2024 - May 2025
MARMot Lab, National University of Singapore | Dr. Guillaume A. Sartoretti — *Singapore*

- Developed a versatile imitation learning framework integrating expert demonstrations directly into RL, grounding behavior with expert-informed priors while maintaining exploration efficiency.
- Developed an industrial grade automated robotic locomotion and manipulation framework using central pattern generators for deploying twist-locks in container yards as part of a project in collaboration with the Port of Singapore (PSA Corporation Limited) and Singapore Ministry of Education (MOE).
- Implemented DRL parallel training algorithms for legged robots (hexapods/quadrupeds), achieving successful sim-to-real transitions and hardware validation.
- Researched on adaptive curriculum reinforcement learning via Unsupervised Environment Generation to increase training efficiency with reduction of training time reducing the sim-to-real gap.

PUBLICATIONS

- **APEX: Action Priors Enable Efficient Exploration for Robust Motion Tracking on Legged Robots**
  S. Sood*, L. B. Nakhwa*, Y. Cao, S. Ge, J. Cheng, F. Zargarbashi, T. Yoon, S. Choi, S. Coros, G. Sartoretti
  Joint research: National University of Singapore (NUS), ETH Zurich, and Korea University
  Under Review, 2026 (*Equal contribution)
  [Paper] · [Project Website]

CONFERENCE

- **Imitation Beyond Data: Scaffolding Reinforcement Learning with Action Priors from Nature**.
  Spotlight Oral Presentation 21st Dynamic Walking Conference, Aachen, Germany, July 2025

PROJECTS

SpiderBot: Led a 25-student team to build a 6-legged hexapod from scratch. Deployed a Bio-inspired Central Pattern Generator (CPG) controller for adaptive gait generation. *Repository*

Predicting Autoimmune Diseases: Developed predictive models for PIBD using Random Forest and PCA to identify significant genes in Rheumatoid Arthritis and Sjogren's Syndrome. *Repository*

Motion Tracking for Yoga: Led development of a real-time pose tracking system using LiDAR, depth cameras, and MediaPipe, incorporating automated corrective feedback algorithms. *Repository*

AWARDS

- Secured second place in the Open Design Contest 2023, BITS Pilani, Goa Campus.

VOLUNTEERING

- Guided 300+ participants in a workshop on **Autonomous Maze Solving using ROS** (Quark 2022).
- Core Member, Electronics and Robotics Club (3 years); Lab Committee, Sandbox Innovations Lab.